Table of Contents

Revision Log

Author	**Date**	**Comments**
Sebastian Fine	Jul-09-2024	Original Author

REVAMP Version Log

Version	**Date**	**Contributor**
1.0.0.0	Oct-22-2024	Sebastian Fine
1.1.0.0	Feb-10-2025	Sebastian Fine

1. FILE STRUCTURE REQUIREMENTS

1.1 DESCRIPTION

The REVAMP Software requires specific formatting for the .txt files placed into the revampData\RAW file directory. The existence of the revampData\RAW file directory ensures that any data further processed from this point of access will follow consistent behavior for saving resultants of subsequent processes.

1.2 FILE NAMING CONVENTIONS

Files which are to be processed by the REVAMP software must abide by the file naming conventions described below. The naming conventions ARE CASE SENSITIVE!

- **Option 1** – ***5 Underscore Delimiters***
  - subjectIdentifier_operatorIdentifier_experimentName_recordingExperiment_recordingDate_recordingTime
    - **subjectIdentifier** – Initials, randomized digits, abbreviations of participant
    - **operatorIdentifier** – Initials, or abbreviation, of recording operator
    - **experimentName** – Abbreviations, agency, or identifier of experiment. ***This identifier will generate the directory to reference any subsequent data and IS CASE SENSITIVE!***
    - **recordingExperiment** – Denotes the subsection of the experiment being recorded.
    - **recordingDate** – Denotes the Gregorian Calendar date of recording
    - **recordingTime** – Denotes the 12-Hour timestamp for recording
- **Option 2** – ***6 Underscore Delimiters***
  - subjectIdentifier_operatorIdentifier_experimentName_recordingExperiment1_recordingExperiment2_recordingDate_recordingTime
    - **subjectIdentifier** – Initials, randomized digits, abbreviations of participant
    - **operatorIdentifier** – Initials, or abbreviation, of recording operator
    - **experimentName** – Abbreviations, agency, or identifier of experiment. ***This identifier will generate the directory to reference any subsequent data and IS CASE SENSITIVE!***
    - **recordingExperiment1 & recordingExperiment2** – Denotes the subsection of the experiment being recorded ***and will be concatenated to generate a directory based on the resultant string***.
    - **recordingDate** – Denotes the Gregorian Calendar date of recording
    - **recordingTime** – Denotes the 12-Hour timestamp for recording

1.3 FILE STRUCTURE CONVENTIONS

The internal structure of recorded data contained within ANY .txt file attempting to be processed utilizing REVAMP must abide to the following structural conventions. These conventions ARE CASE SENSITIVE! The requirements for each file's internal structure are provided below.

- **Keywords (Abbreviated File Structure Overview)**
  The minimum keywords required for a singular movement recording
  - Section Header
    - Collected

            - Right Eye Horizontal
            - Right Eye Vertical
            - Right Eye Pupil
            - Left Eye Horizontal
            - Left Eye Vertical
            - Left Eye Pupil
        - End Trial
- **<u>Section Header</u>**
    - Section Headers must be present above the keyword “Collected” to allow REVAMP to successfully identify the onset of a movement recording.
    - Section Headers must be unique for DISTINCT stimulation recordings. Section Headers which are identical will place all observations of the specified Header into a singular collection.
- **<u>Data Channels</u>**
    - Data Channels are defined by their RIGHT and LEFT Eye recordings. All Channels must be present.

2. INTERFACES

2.1. REVAMP MAIN MODULE

I. DESCRIPTION

The REVAMP MAIN MODULE is a container which provides a user with the ability to

- IMPORT raw .txt files into the REVAMP software
- SELECT AND FILTER DATA
- Enter the CALIBRATION MODULE
- Enter the ANALYSIS MODULE.

2.2. IMPORT DATA

I. DESCRIPTION

The IMPORT DATA Module is accessible via the “Import Data” button within REVAMP MAIN MODULE. When pressed, a file directory window will appear. The default pathing for this directory searches for revampData\RAW, however the user can navigate to a secondary directory. All subsequent processing will occur within the specified root directory selected by the user.

II. WORKFLOW

The generalized workflow for importing .txt file data into REVAMP is as follows:

1) ImportDataButtonPushed
   a. revampPreprocessor()
      i. importRawData()
      ii. saveFunction()

2.3. FILTER DATA

I. DESCRIPTION

The FILTER DATA module is accessible via the “Filter Data” button within REVAMP MAIN MODULE. When files are previously selected via IMPORT DATA, the Filter Data “Data Available” Lamp will illuminate following successful importing. When files are not previously selected, the “Select Data” Button within the “Filter and Calibrate Data” container must be utilized by the user to select data for filtering. User-selected data MUST be from the UNFILTERED_UNCALIBRATED directory!

II. WORKFLOW

The generalized workflow for filtering UNFILTERED_UNCALIBRATED data files is as follows:

1) filterTable.filterTableData()
2) saveFunction()

III. BLINK REMOVAL

1) DESCRIPTION

The BLINK REMOVAL feature is contained within the FILTER DATA Module and requires TWO user-inputs for Blink Handling. BLINK REMOVAL is calculated by the identification of outliers via “gesd” in the vertical AND pupil trace. Thresholds for identified outliers are values observed to be 2.5 greater than the median recorded value of the file for vertical, and .5 greater then the median recorded value of the file for pupil.

- Sample Threshold (Samples) – Specifies the number of samples to set in determining if a “Blink in Transient” occurs. (Default: 750)
- Blink Fill Method (String) – Specifies the interpolation method for filling Blink Data. (Default: “previous”).

IV. KEY VALUES

1) DESCRIPTION

The key values, and their default values for the FILTER DATA Module are as follows:

- Vergence Primary Cut-Off Frequency: (Default: 40Hz). Provides the “low” and “high” fc value for filtering.
- Vergence Secondary Cut-Off Frequency: (Default: 0Hz). Provides the secondary fc for “bandpass” filter configurations.
- Pupil Cut-Off Frequency: (Default: 5Hz). Provides the primary fc for “low” pass filtering.
- Saccade Primary Cut-Off Frequency: (Default: 100Hz). Provides the primary fc value for “low” and “high” pass filtering.
- Saccade Secondary Cut-Off Frequency: (Default: 0Hz). Provides the secondary fc value for “bandpass” filtering.
- Filter Type: (Default: “low”). Provides the filtering configuration passed to MATLAB “butter” filter coefficient generator function.
- Filter Order (Default: 3). Provides the order of the filter to be created.
- Sampling Rate (Default: 500Hz). Provides the sampling rate of the data being filtered.

2.4. CALIBRATE DATA

I. DESCRIPTION

The CALIBRATE DATA Module is accessible through the CALIBRATE DATA Button within the REVAMP MAIN MODULE. Upon interacting with the CALIBRATE DATA Button a secondary calibrationModule window will appear. Within the calibrationModule, users can utilize the LOAD DATA Button to select data from the FILTERED_UNCALIBRATED directory.

Key Aspects of the calibrationModule:

- Pupil Calibration GAINS are specified within the REVAMP MAIN MODULE and contain the outputted values from ISCAN recording software.
- Calibrations are identified by the recorded “Section Header” containing the keyword “CAL”.
- Calibration Type is specified by the “Horizontal”, “Vertical”, and “Pupil” Data Channels.
- Calibration regression and gain are obtained by the MEAN value for each calibration Section for LEFT AND RIGHT!
- Left (and Right) Eye Gain within the “Regression Information” container is a spinner and accepts ANY numerical data (including negative values).
- “Gain Evaluation” container provides the user with the ability to specify the plotted movement within “Movement Selection” and returns the MEAN recorded value between START INDEX and END INDEX.

- CALIBRATION of the “Right” and “Left” eye is specific to each “Calibration Type”. Hence to calibrate Horizontal, Vertical, and Pupil each section should be selected “Calibration Type”.

2.5. ANALYSIS

I. STEP ANALYSIS

i. DESCRIPTION

The STEP ANALYSIS Module contains an omnibus approach to analysis of Horizontal, Vertical, and Pupil movements. The LOAD DATA Button requires files be selected from the FILTERED_CALIBRATED directory. The Movement Ensemble, Velocity Ensemble, Individual Movement, and Individual Velocity Axes will populate once data is selected by the user. Metric Analysis for each Data Channel is performed by the "Eye Selector" and "Data Selector".

Section Headers are navigated via the "Section Name" dropdown. Classification is simplified to the basis of analyzable (Good) or not-analyzable (Bad) with a Classification Comments section which has preloaded comments (Default: "None") and the ability for the user to type in a specific comment string for EACH EYE CHANNEL. Plotting behavior is specified by "Summation" akin to Vergence, and "Average" which is akin to Version. The deviation of this naming convention is due to the variations in behavior of Pupil (Average Only) responses and their Horizontal (Summation or Average) movement type.

Individual plots can be exported via the Save Individuals Button, or Movement and Velocity ensembles via the Save Ensembles Button. "Scale Ensemble Velocity to Analysis" Checkbox enables REVAMP to apply a gain to the calculated Ensemble Velocity which is consistent with the current average metric analysis.

Individual Movements within each Section Header can be navigated via the "PREVIOUS" and "NEXT" Button. Individual Files can be navigated via the "Previous File" and "Next File".

The "Save" button will save the current metric table for the TOTAL file – containing all section headers, movements, and classifications of each eye and data channel.

II. FFT ANALYSIS

i. DESCRIPTION

III. EXPORT ANALYSIS

ii. DESCRIPTION

EXPORT ANALYSIS is a button within the analysisModule, which allows the user to identify ANALYSIS_METRICS. Selected data from the user will be compiled with file-wise BLINK_DATA which contain BLINK_COUNT, BLINK_IN_TRANSIENT, and BLINK_INDEX information per file and per movement within each Section Header. Compiled exports are saved as "DataExports" with a unique UNIX timestamp to protect from accidentally overwrites.

3. HOT KEYS & ACRONYMS

3.1. REVAMP MAIN MODULE

- "i"
  - Execute importDataButtonPushed
- "f"
  - Execute filterDataButtonPushed
- "c"
  - Open calibrationModule
- "a"
  - Open analysisModule
- ESCAPE KEY
  - Exist REVAMP MODULE

3.2. CALIBRATE DATA

- "a"
  - Load PREVIOUS file
- "d"
  - Load NEXT file
- "s"
  - SAVE data
- "r"
  - Calibrate RIGHT eye
- "l"
  - Calibrate LEFT eye
- RIGHT ARROW KEY
  - Change plotting behavior to AVERAGE
- LEFT ARROW KEY
  - Change plotting behavior to SUMMATION
- "h"
  - Change calibration type to HORIZONTAL
- "v"
  - Change calibration type to VERTICAL
- "p"
  - Change calibration type to PUPIL

3.3. ANALYSIS

I. STEP ANALYSIS

- Load Data
  - "l" – Load Data
- Plotting Behaviors
  - RIGHT ARROW KEY – Plot AVERAGE of data
  - LEFT ARROW KEY – Plot SUMMATION of data
  - "n" – Activate NULLIFY tool
  - "y" – Activate Scaled Velocity Checkbox
- Metric Analysis
  - "q" – Select RSI (Latency)

        - "w" – Select Peak Velocity Button
        - "e" – Select REI (End of Transient)
        - "r" – Calculate ID
        - "t" – Calculate Final Amplitude
    - File Navigation
        - "z" – Load PREVIOUS file
        - "c" – Load NEXT file
    - Movement Navigation
        - "a" – Load PREVIOUS movement
        - "d" – Load NEXT movement
        - DOWN ARROW KEY – Load NEXT SECTION below
        - UP ARROW KEY – Load PREVIOUS SECTION above
    - Saving Behaviors
        - "f" – Save Individual Movement and Velocity plots
        - "v" – Save Ensemble Movement and Velocity plots
    - Data Channel Selection
        - "h" – Select "Horizontal" Data Channel
        - "j" – Select "Vertical" Data Channel
        - "k" – Select "Pupil" Data Channel
    - Eye Selection
        - "1" – Select "Left Eye"
        - "2" – Select "Right Eye"
        - "3" –Select "Binocular"
    - Classification
        - "g" – Set Classification as "Good"
        - "b" – Set Classification as "Bad"
    - Closing Command
        - ESCAPE KEY – Prompt Exit of analysisModule
    - Save Command
        - "s" – Save Metric Table for FILE

II. ACRONYMS

- RSI – "Response Starting Index" akin to Latency, or the onset of the movement where the velocity initially deviates from the X-Axis
- REI – "Response Ending Index", the offset of the movement or ending of the movement response where the velocity re-intercepts with the X-Axis
- RA – "Response Amplitude", calculated by the positional difference of where REI and RSI are analyzed to be: x(REI) – x(RSI)
- PVI – "Peak Velocity Index" the temporal position of the analyzed peak velocity
- PV – "Peak Velocity" the positional value of where PVI is selected
- FA – "Final Amplitude" the positional AVERAGE of the last .5 seconds of recording
- RSID – "Response Starting Index Diameter", Pupil Response variation of RSI
- REID – "Response Ending Index Diameter", Pupil Ending Index variation of REI

- RAD – "Response Amplitude Diameter", Pupil Diameter difference from RSID to REID where: x(REID) – x(RSID)
- PDROCI – "Pupil Diameter Rate of Change Index", temporal Pupil Peak Velocity Index
- PDROC – "Pupil Diameter Rate of Change", positional Pupil Peak Velocity
- FD – "Final Diameter" the positional AVERAGE of the last .5 second of recording
- ID – "Initial Diameter" the positional AVERAGE of the first .5 second of recording
- HClass – "Horizontal Classification"
- VClass – "Vertical Classification"
- PClass – "Pupil Classification"